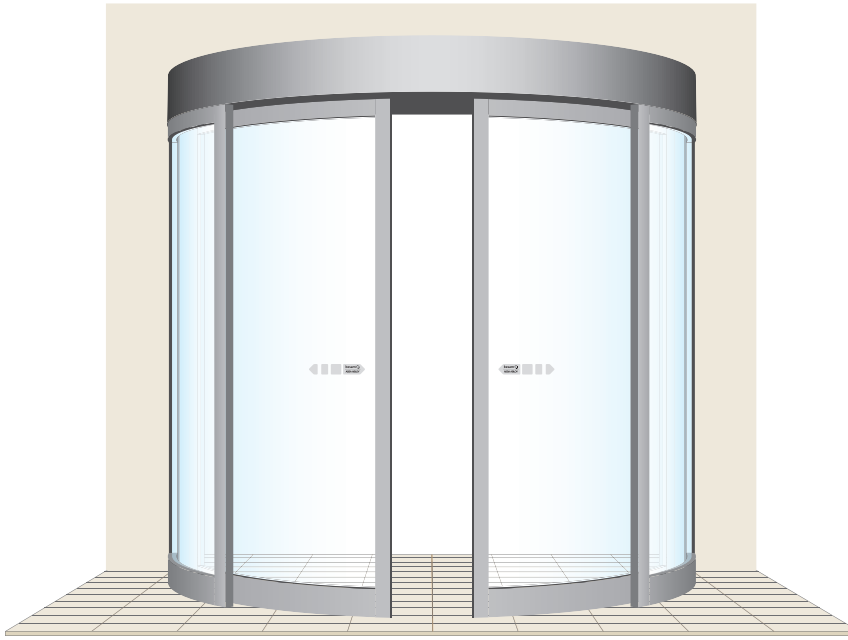
Besam Curved Sliding Door System CMD

User Manual

Original instructions



ILL-00224

CONTENTS

Congratulations on your new automatic door! 3
Electronic equipment reception interference 3
Environmental requirements 4
Product liability 4
Warranty 4
Service 5
Purpose of use 6
How the CMD works 6
Programme Selectors 8
Technical specification 9
Standard safety 10
Optional safety 11
Emergency escape 11
Regular safety checks 12
Safety accessories 15
General accessories 15
Troubleshooting 17
Maintenance plan 18
Other products from Besam 18
Declaration of conformity 19

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Backtrack information: folder: Library, version: a172, date: 2011-05-03, time: 08:17:04, state: Working

Congratulations on your new automatic door!

Besam has developed automatic doors for more than 40 years. State-of-the-art technology and carefully tested materials and components provide you with a superior product.

As with all other technical products, your automatic door requires periodic maintenance and service. This manual will assist you with this. It is essential that you know your automatic door (system) and that you recognize the importance of maintaining it in compliance with applicable standards for safety.

Your local Besam-authorized representative is familiar with these standards, as well as all applicable local codes and Besam recommendations for power-operated pedestrian doors. Service and adjustments performed by your Besam-authorized representative, will ensure safe and proper operation of your automatic door unit.

HAZARD WARNING!



Failure to observe the information in this manual may result in personal injury or damage to equipment.

Save these instructions for future reference.

Electronic equipment reception interference

This equipment may generate and use radio frequency energy and if not installed and used properly, that is, in strict accordance with the manufacturer's instructions, it may cause interference to radio, television reception or other radio frequency type systems. It has been designed to comply with the emission limits in accordance with EN 61000-6-3 (US market FCC 47 CFR part 15 B), which are designed to provide reasonable protection against such interference in a residential installation.

However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- Re-orient the receiving antenna.
- Relocate the receiver with respect to the equipment.
- Move the receiver away from the equipment.
- Plug the receiver into a different outlet so that equipment and receiver are on different branch circuits.
- Check that protective earth (PE) is connected.

If necessary, the user should consult the dealer or an experienced electronic technician for additional suggestions.

Environmental requirements

Please act according to your local regulations and dispose of your old product(s) and packaging properly. The correct disposal will help prevent potential negative consequences for the environment and human health.

Besam products are equipped with electronics and may also be equipped with batteries containing materials which are hazardous to the environment. Remove this material from the operator before it is scrapped and make sure that it is disposed of properly as was done with the packaging.

Product liability

According to regulations, the following are the responsibility of the owner or caretaker of the equipment

- that the equipment operates correctly, so that it gives sufficient protection in regard to safety and health
- that the equipment is operated and regularly maintained, inspected and serviced by someone with documented competence in the equipment and in applicable regulations
- that a journal is kept of inspection, maintenance and service records including the provider
- that inspection covers the emergency opening function (when applicable)
- that the closing force is appropriate for the door size on fire-approved door systems (when applicable)

Warranty

Besam warrants its products to be free from defects in material and workmanship under intended use and service for a warranty time of 12 months, beginning at time of delivery. This warranty extends only to the original buyer of the equipment.

Besam warrants that the software will operate substantially in accordance with its functional descriptions and that it has been recorded on non-defective media.

The Besam warranty does not apply to

- That the software will be error-free or operate without interruption
- General wear and tear on the system
- Fuse, disposable batteries and glass damage
- System deviations caused by installer other than Besam
- System that has been altered or damaged by vandalism or misuse
- System that has been additionally equipped with non-Besam original branded parts and/or spare parts
- Unrequired visits due to poor client communication (door working when our technician arrives, reset, power discontinuation)
- Adjustments (closing and opening speed and also detection field radars) due to customer requests
- Water damage
- Adverse weather conditions
- Any damage caused, directly or indirectly, by a circumstance beyond the control of the applicable company within ASSA ABLOY Entrance Systems AB, such as industrial dispute, fire, natural disaster, war, extensive military mobilization, insurrection, requisition, seizure,

embargo, restrictions in the use of power and defects or delays in deliveries by sub-contractors caused by any such circumstances

Please note:

- Non-compliance with manufacturers care and maintenance recommendations may void the warranty.
- Besam -authorized resellers shall extend this warranty to end-users only, but have no authority to extend a greater or different warranty on behalf of Besam.
- A service agreement with Besam will secure the availability of a fully operational system and will give priority at call-out, thus minimizing the time that the system is unfit for use.

Service

Regular inspections by a trained and qualified person, and the frequency of service occasions, should at minimum be made according to national regulations. This is especially important when the installation concerns a fire-approved door or a door with an emergency-opening function. To extend the life of your investment and ensure safe and reliable operation of the door, we recommend a minimum of 2 visits per year or more, depending on usage and operating conditions. Talk to your Besam representative for the best solution for your installation and to learn more about Besam Pro-Active Care! For contact information, see last page of this manual.

Purpose of use

The CMD is an automatic sliding door operator developed to make life easier for everyone who can benefit from automatic doors. The CMD is designed to offer continuous use, a high degree of safety and maximum lifetime.

Besam CMD sliding door makes it possible to design the entrance in three dimensions, with depth in addition to the normal width and height. This gives the entrance an elegant and decorative appearance whilst it remains extremely functional. A curved sliding door gives a wider opening than a flat sliding door.

The system is self-adjusting to the effects caused by normal variations in the weather conditions and to minor friction changes caused by e.g. dirt.

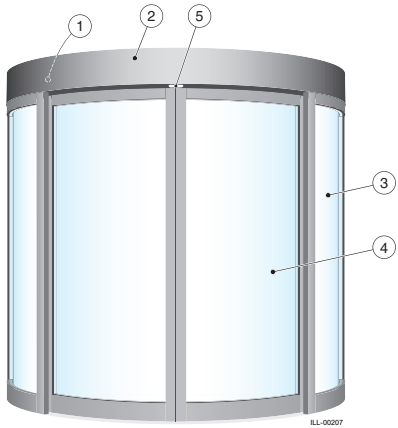
Each installation is, however, unique and must therefore be equipped and adjusted for the application-relevant safety requirements, just like maintenance must be performed as specified for the selected product in a given environment.

How the CMD works

The Besam CMD sliding door system works electromechanically. The drive unit, control unit, transmission, optional emergency unit and electromechanical locking device are all assembled in a support beam with an integrated cover. The motor and gear box transmit movement to the door leaves by means of a tooth belt. The door leaf is fitted to a door adapter/carriage and hangs on a sliding track. The active leaf travel is directed by a floor guide.

When an opening impulse (e.g. from a motion sensor) is received by the control unit, the drive unit starts and transmits movement to the door leaves to open.

The closing starts when no OPENING IMPULSE is received, the HOLD OPEN TIME has ended and no PRESENCE IMPULSE is activated.



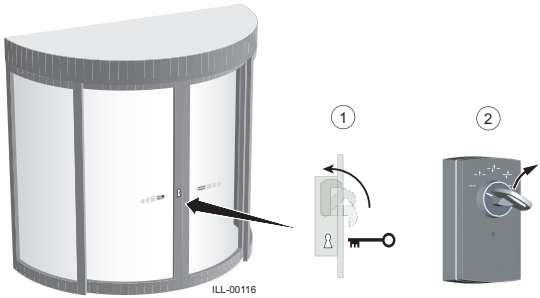
1 Operator cover
2 Operator support beam (incl. sliding track)
3 Screens left and right
4 Doors left and right (incl. floor guide)
5 Presence sensor

Locking

Doors used for emergency escape in buildings such as hospitals and homes for elderly people may not be locked or put in programme selection OFF. In other buildings emergency escape doors may be locked or put in programme selection OFF after it is secured that all people has left the building.

Unlocking

Unlock all the mechanical locks before switching on the operator. Note that more than one lock may be fitted.



Programme Selectors

The CMD user can select different modes of operation if a programme selector is installed. See the table below.

- PSMB-5, for control of one operator only.
- PS-5M, for central control of an optional number of operators. In setting AUTO every connected operator are individually controlled by its own programme selector. Reset is not available on PS-5M.
- PSK-6U, surface mounted on the side screen or on the wall close to the door.

Programme selector functions

	OFF	This function is only used on emergency escape doors after it is certain that all people have left the building. The door cannot be opened with inner and outer activation units. The door is locked if an electromechanical locking device has been fitted. The door can be opened partially with a key switch (if fitted). With an emergency push-button (if fitted) the door opens fully.
	EXIT	Passage from inside only. The door is normally locked if an electromechanical locking device has been fitted. The door can only be opened with the inner activation unit or with a key switch/emergency push-button (if fitted).
	AUTO	Two-way traffic, normal operation of the door. The door can be opened with the inner and outer activation units and with a key switch/emergency push-button (if fitted).
	AUTO PARTIAL	Two-way traffic, AUTO PARTIAL is obtained. The door can be opened partially with the inner and outer activation units and with a key switch (if fitted). With an emergency push-button the door opens fully.
	OPEN	The door is permanently open. The door can be moved by hand e.g. for window cleaning. All activation units except for the emergency push button (if fitted) are disconnected.
	RESET	Set the programme selector to AUTO. Insert a narrow object in the small hole on the programme selector and push briefly. The operator makes a system test of the emergency unit (if selected), electromechanical lock, watchdog relay and closed door position. The operator is after closing reset and ready for normal operation. **Note!** The RESET is not available on programme selector PS-5M.



RESET Turn the key clockwise to the position “R” (six o’clock) and insert a narrow object in the small hole on the programme selector and push briefly. Then turn the key counter-clockwise back to the requested setting. The operator makes a system test of the emergency unit (if selected), electromechanical lock, watchdog relay and closed door position. The operator is after closing reset and ready for normal operation.

Note! The key cannot be removed in the "R" position.

Note! If monitored emergency unit is a demand, a test of the emergency unit is performed when the programme selector is turned from OFF or OPEN to any other position.

Note! If monitored emergency unit is a demand, a test of the emergency unit is performed when the programme selector is turned from OFF or OPEN to any other position.

Technical specification

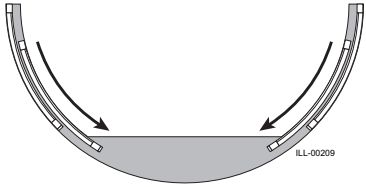
Mains power supply: 120 V AC -10% to 240 V AC +10%, 50/60 Hz, fuse 10 AT

Power consumption: Max. 250 W

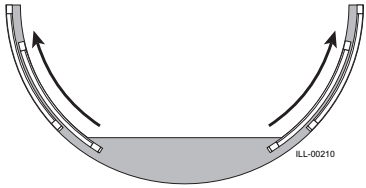
Third party approvals from established certification organizations valid for safety in use and escape route safety. For details see Declaration of conformity.

Standard safety

To permit a safe passage between closing doors, the doors reverse immediately if an obstruction is met. They then resume their interrupted movement at low speed to check whether the obstruction has disappeared or not.



If an obstruction is met between opening doors and surrounding walls or interior fittings, the doors stop immediately, and then close after a time delay.



Microprocessor for precise control

The microprocessor has a routine for self-monitoring, which detects any interference or faulty signals in door operation. If an input signal does not correspond to the preprogramming, the microprocessor automatically takes necessary actions to ensure a safe operation.

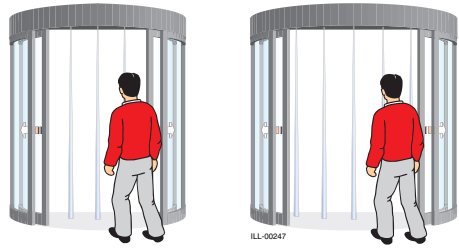
Safety system with presence sensor PDR

The safety system incorporates a presence sensor installed above the door opening. If the presence sensor detects an object in the doorway, while the doors are closing, the doors reverse immediately. They then resume their interrupted movement as soon as the object is removed.



Optional safety

Two or four **extra** safety presence sensors can be included. The number of extra sensors are depending on the opening width. The distance between the photocells is 400 mm.



Emergency escape

The CMD can be combined with an emergency unit that automatically opens or closes[1] the doors in the event of a power failure and can also be interfaced with the fire alarm or smoke detector. For further information, see .

Doors used for emergency escape in buildings such as hospitals and homes for elderly people may not be locked or put in programme selection OFF.

[1] Electrical emergency unit only

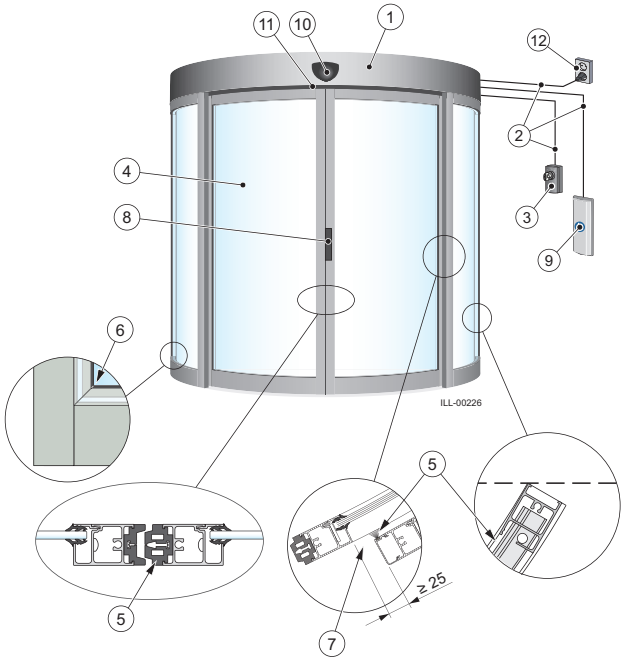
Regular safety checks

To help you fulfil the national/international requirements and to avoid malfunction and risk for injuries, we have provided the following checklist.

Do not use if repair or adjustment is necessary.

Disconnect supply ⑫ when cleaning or other maintenance is to be carried out.

FUNCTION AND VISUAL INSPECTION



CLEANING

The best way to remove dust and dirt from the CMD is to use water and a soft cloth or a sponge. A gentle detergent may be used. To maintain the quality of the enamel layer, the surfaces should be cleaned once/four months period. The cleaning should be documented. To avoid damages to the profiles the brushes/weather stripping must be vacuum-cleaned weekly.

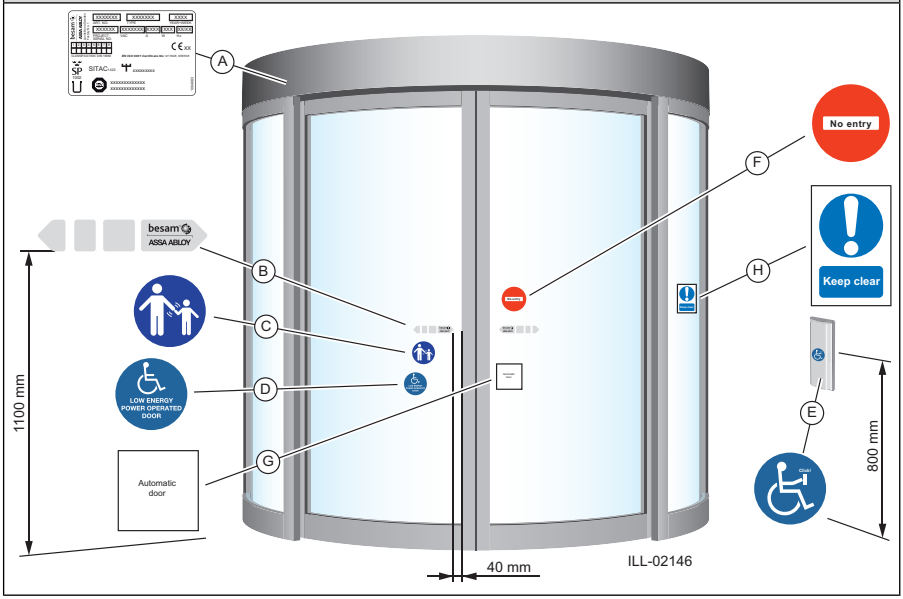
- Do not expose windows, doors or profiles to alkalis. Both aluminium and glass are sensitive to alkalis.
- Do not clean with high pressure water. Operator, programme selector and sensor may be damaged and water may enter the profiles.
- Do not use polishing detergent.
- Do not scrub with materials like Scotch-brite, as this will cause mechanical damage.

Action			If problem occurs
Activate your operator and *visually* check, fastening of • operator and cover ① • cables ② • programme selector(s) ③ • door and glass (stability) ④			
Also inspect your operator and check *visually* for • condition of door seals and weather stripping ⑤ • condition of glazing rubbers ⑥ • finger protection ⑦ • proper operation; closes slowly and smoothly • any ventilation being obstructed			
Set the programme selector to OFF and check that the operator and electromechanical lock (if fitted) work together. Also check that the lock really secures the door ⑧.			
Activate the manual activation units ⑨, if any, and walk towards the door. Check that the door has opened sufficiently while you pass the entrance/exit. Then proceed with the automatic activation units ⑩ in the same way.			
Check the safety sensors ⑪ if any, see page 10. If you are unsure of which type of sensor you have, please contact your Besam representative.			
Escape doors	By law, these tests must be performed regularly by trained personnel.	If the operator is equipped with automatic opening system, shut off the power and the door should open and remain open. Restore power and the door should close.	

= Contact your Besam representative. For contact information, see last page.

= Take appropriate measures.

SIGNAGE



ACTION		If problem occurs
Check that all required signage is applied and intact. Mandatory indicates that the signage is required by European directives and equivalent national legislation outside the European Union.		
Ⓐ	Product label: Mandatory	
Ⓑ	Besam door sticker: Mandatory, if applicable to highlight the presence of the glass (applied to all glass sections that are moving).	
Ⓒ	Supervision of child: Mandatory, if applicable (applied to both sides of the door). To be placed on entrances where the risk analysis shows use by children, elderly and disabled.	
Ⓓ	Operator designed for disabled people: Recommended, if applicable (applied to both sides of the door)	
Ⓔ	Activation by disabled people: Recommended, if applicable	
Ⓕ	No entry, identifying one-way traffic: Mandatory in GB and US, if applicable	
Ⓖ	Automatic door: Only mandatory in GB	
Ⓗ	Keep clear: Only mandatory in GB	

= Contact your Besam service representative. For contact information, see last page.

Safety accessories

Even though the CMD is installed to comply with all applicable safety regulations, it is possible to enhance safety/comfort with the following add-ons (please contact your local Besam company for detailed description).

- Combined motion and presence sensors
- Separate presence sensors

General accessories

Your CMD can be further improved with the following add-ons (please contact your local Besam company for detailed description).

- Cover

  Made in clear anodized aluminium as standard. Paint finished in RAL colours or anodizing optional.
- Motion and presence sensors, see separate manuals or installation drawings.
- Programme selectors
  (see page 8)
- Programme selector PS-2
- Electrical locks

  Locked with power, locked without power or bistable lock.
- Micro switch kit

  For indication of door and lock position.
- Electrical emergency unit with batteries

  Used if a door is required to be opened or closed by means of a rechargeable battery unit and remain in this position in the event of power failure. Authorities can demand that the emergency unit is monitored on a regular time basis. Half an hour before this time has elapsed the following opening impulse generates an emergency opening test. If there is no opening impulse within an additional time, specified by the authorities, the operator control unit generates the opening impulse itself.

  If the battery opens the door within the limited time the test is successful and the door resumes the function set by the programme selector.

  Note! The test is never performed in programme selector setting OPEN. In setting OFF it can be selected. The test is always performed after a RESET and after changing program selection, from a position where a test is not done to a position where the test is a demand.
- Emergency closing with repeated closing

  If the door is opened by hand after an electrical emergency closing, it will close again.
- Mechanical emergency unit with elastic cord

  Used if a door is required to be opened and remain opened by means of an elastic cord in the event of power failure. Authorities can demand that the emergency unit is monitored on a regular time basis. Half an hour before this time has elapsed the following opening impulse generates an emergency opening test. If there is no opening impulse within an additional time, specified by the authorities, the operator control unit generates the opening impulse itself.

  If the elastic cord opens the door within the limited time the test is successful and the door resumes the function set by the programme selector.

Note! The test is never performed in programme selector setting OPEN. In setting OFF it can be selected. The test is always performed after a Reset and after changing program selection, from a position where a test is not done to a position where the test is a demand.

- Interlocking

  Used between two operators when the first operator must close before the other one can open (typical to reduce energy losses and not for security reasons).
- Convenience battery UPS

  Stand-by supply which gives continued operation during short power failure.
- Emergency opening

  Opens the door in any programme selector setting (fireman's opening).
- External error indication

  Obtained if a lamp or a buzzer is connected.
- Key switches (flush and/or surface mounted)

  Used to give opening impulse to the door in any programme selector setting. With electrical emergency unit also during power failure.
- Push button

  Used to give opening impulse to the door.

Troubleshooting

What's wrong	Remedies
The door does not open	
The motor does not start	Change the setting of the programme selector.
	Check the mains switch.
The motor starts but stops during opening	Unlock the mechanical locks.
	Clean the floor guide.
	Check for objects jammed under the door.
The door does not close	
The motor does not start	Change the setting of the programme selector.
	If a presence sensor is installed, check for and remove objects placed in the presence zone.
The motor starts but stops during closing	Clean the floor guide.
	Check for objects jammed under the door.
The door moves slowly	
	Prevent traffic using the door and allow it to do a complete opening and closing cycle with low speed.
	Reset the operator by setting the programme selector to AUTO and pushing a narrow object in the small hole on the programme selector.
	Allow the operator to control the closed position without interruption.
If the problem continues, please contact your Besam representative.	

Maintenance plan

As mentioned earlier in this manual, service and adjustments performed by your Besam-authorized representative will ensure safe and proper operation of your automatic door unit. The table below shows the recommended interval in months, when to replace parts during preventive maintenance. Talk to your Besam representative to learn more about Besam Pro-Active Care!

Part	Part Number	Cycles/hour in operation			Abusive Environment
		<10	<100	>100	
		Low Traffic	Medium Traffic	High Traffic	
Electrical emergency unit battery	33550475	24	24	24	24
Mechanical emergency unit elastic cord	331700121	12	12	12	12
Floor guide shoe Felt padded	33831622	24	12	6	6
Carriage wheel Single Anti-derailing	830060 246313	36	24	12	12
Tooth belt	33735251	60	48	36	24

Check that all required signage is applied and intact, see page 14. Also check other consumable parts, such as brushes, door stops and glazing rubbers.

Other products from Besam

- Door Systems
- Balanced doors
- Air curtains
- Rollershutters
- Revolving doors
- Sliding doors
- Swing doors
- Automatic and manual activation units
- Service such as preventive maintenance and upgrade programs, emergency repairs, service advice and door management

Declaration of conformity

We **ASSA ABLOY Entrance Systems AB**
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Sweden

Telephone **+46 418 511 00**
Telefax **+46 418 511 32**
E-mail **general@assaabloyes.com**

declare under our sole responsibility that the type of equipment:
CMD. With or without emergency unit.

complies with the following statements.

EC directive applied to the product:

A	X	Low Voltage Directive 2006/95/EC
B	X	ElectroMagnetic Compatibility Directive 2004/108/EC
C		Construction Product Directive 89/106/EEC + 93/68/EEC
D	X	Machinery Directive 2006/42/EC

Harmonized European standards which have been applied:
EN 60204-1 EN 60335-1 EN 61000-6-2 EN 61000-6-3 EN ISO 12100-1/-2
EN ISO 14121-1

Other standards or technical specifications, which have been applied:
BBR BGR 232 BVL DIN 18650-1/-2
UL 325

EC type examination or certificate issued by a notified or competent body (for full address, please contact ASSA ABLOY Entrance Systems AB) concerning the equipment:
SITAC 0002/01 SITAC 0138/01
TÜV B 10 04 58029 012 UL File E47833

A	B		D

The equipment conforms completely to applicable parts of stated harmonized standards and

			D

The equipment conforms wholly or partially with other standards or technical specifications and to state of the art and technology.

The manufacturing process ensures the compliance of the equipment with the technical file. The manufacturing process is regularly accessed by 3rd party.

The last two digits of the year in which the CE marking was affixed:

A 10 B 10 C -- D 10

Place	Date	Signature	Compilation of technical file
Landskrona	2010-06-09	Johnny Persson	Address as above

Position
Product Safety & IP Manager

701362-EI-4.0

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